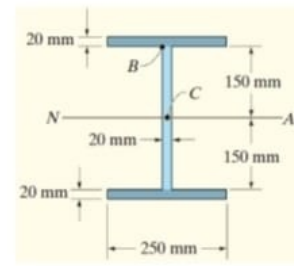
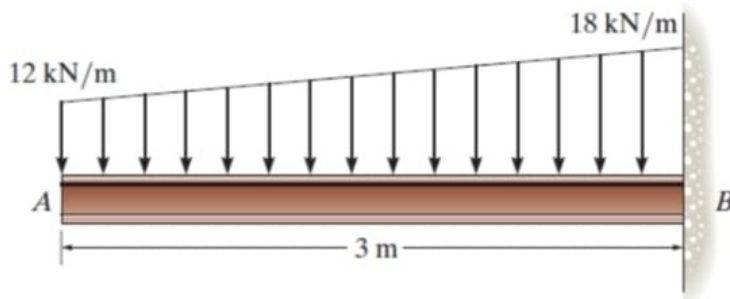
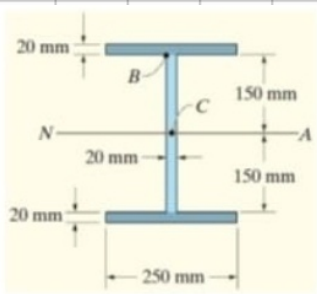


Problem 14: - The cantilever beam in the figure is made from a ductile material and is statically loaded. Determine the safety factor for the critical stress element using appropriate failure theory. (Assume $\sigma_{yt} = 250 \text{ MPa}$).

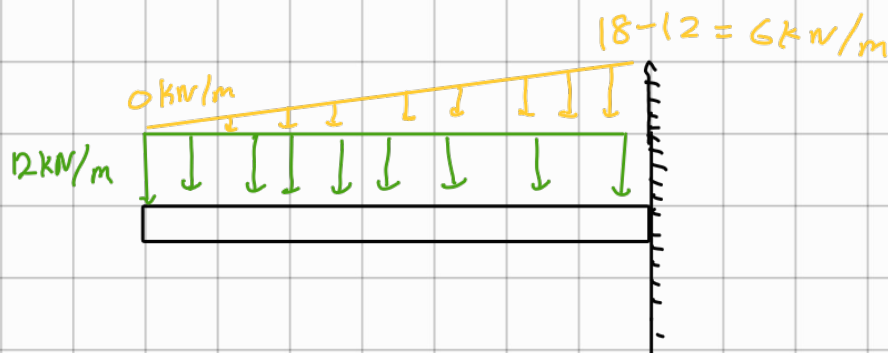


Ans:



$$I_{zz} = \frac{1}{12} \times 20 \times 300^3 + \left(\frac{1}{12} \times 20^3 \times 250 + 20 \times 250 \times 160^2 \right) \times 2$$

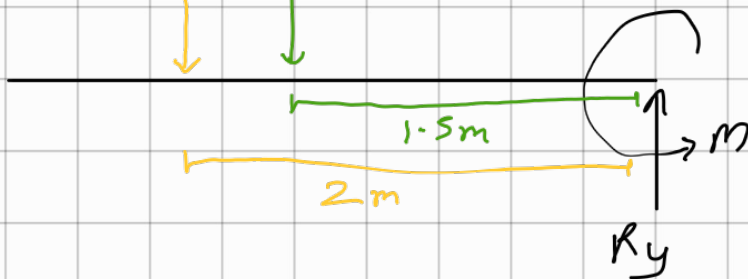
$$I_{zz} = 3.0133 \times 10^{-4} \text{ m}^4$$



FBD

$$F_2 = \frac{1}{2} \times 6 \times 3 = 9 \text{ kN}$$

$$F_1 = 12 \times 3 = 36 \text{ kN}$$



$$M_B = 12 \times 3 \times 1.5 + \frac{1}{2} \times 6 \times 3 \times 2$$

Bending moment at B is max because of free end.

$$= 72 \times 10^6 \text{ N-mm}$$

Shear Force (max)

$$R_y = V_{\max} = 12 \times 3 + (0.5 \times 6 \times 3) = 45 \times 10^3 \text{ N}$$

At point B

$$y = \frac{h}{2} = 170 = 0.17 \text{ m}$$

Bending stress (σ)

$$\sigma_B = \frac{M_{\max} y}{I} = \frac{72 \times 10^6 \times 170}{3.6133 \times 10^8} = 40.62 \text{ MPa}$$

Shear stress

$$\tau_B = 0$$

Point C

$$y = 0$$

Bending stress

$$\sigma_C = 0$$

Shear stress

$$Q = (250 \times 20) \times (150 + 10) + (150 \times 20) \times 75$$

$$\tau_C = \frac{V_{\max} Q}{I \times t_w} = \frac{45 \times 10^3 \times 1.025 \times 10^6}{3.6133 \times 10^8 \times 20} = 7.65 \text{ MPa}$$

Since stress at B > C critical point is B

$$\sigma_1 = 40.62 \text{ MPa} \quad \sigma_2 = 0 \quad \sigma_3 = 0$$

$$\sigma_{yt} = 250 \text{ MPa}$$

$$\sigma_{vm} = \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1 \sigma_2} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{ von Mises.}$$

$$= 40.62 \text{ MPa.}$$

$$n = \frac{\sigma_{yt}}{\sigma_{vm}} = \frac{250}{40.62} = 6.15$$

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{40.62 - 0}{2} = 20.31 \text{ MPa.}$$

Tresca

Theory

$$n = \frac{\sigma_{yt}}{2 \times \tau_{max}} = \frac{250}{40.62} = 6.15$$